

Climatic Risk Intelligence Module

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Submission Documentation

The Climatic Risk Intelligence Module is a Python-based multi-signal decision system that classifies an environment into four safety tiers using temperature, humidity, and wind speed telemetry. The program first validates the raw temperature and humidity inputs using the walrus operator to detect empty values, followed by `try/except` blocks to safely perform floating-point conversion. Humidity is also checked using an assertion because a negative humidity value is physically invalid. The Heat Stress Index (HSI) is calculated using the required formula `temperature + (0.5 * humidity)`. The decision system checks the conditions in the required order: FREEZE ALERT, CRITICAL, CAUTIONARY, and finally OPERATIONAL. Freeze Alert is evaluated first because a freezing temperature must take priority even when the calculated HSI is high. A ternary expression is used to classify the HSI as Safe or Unsafe. The bonus Nested Intelligence Suite is executed only when the initial classification is CAUTIONARY, where the battery level can elevate the status to CRITICAL or downgrade it to OPERATIONAL.

Python Source Code

```
"""
-----
Assignment 2.1 - Climatic Risk Intelligence Module
Course : Programming for Artificial Intelligence
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Description:
This program simulates an AI-based climatic risk intelligence
system. It accepts temperature, humidity, and wind speed
telemetry and calculates the Heat Stress Index (HSI).
```

The environment is classified into four safety tiers:

1. FREEZE ALERT
2. CRITICAL
3. CAUTIONARY
4. OPERATIONAL

The tiers are evaluated in this exact order because Freeze Alert must always take priority over heat-related conditions.

The walrus operator (:=) is used to capture and validate empty input before conversion. The try-except block handles invalid numeric input, while an assertion prevents negative humidity.

The bonus feature asks for battery level only when the initial classification is CAUTIONARY.

```
-----  
"""  
  
print("=====  
Climatic Risk Intelligence Module  
=====\n")  
  
# -----  
# Optional User Name  
# -----  
user_name = input("Name : ").strip() or "Guest User"  
  
# -----  
# Temperature Input  
# -----  
if not (temp_str := input("Temperature (C) : ").strip()):  
    print("Safety Level : Unknown")  
    exit()  
  
try:  
    temperature = float(temp_str)  
except ValueError:  
    print("Safety Level : Unknown")  
    exit()
```

```

# -----
# Humidity Input
# -----
if not (humidity_str := input("Humidity (%) : ").strip()):
    print("Safety Level : Unknown")
    exit()

try:
    humidity = float(humidity_str)
except ValueError:
    print("Safety Level : Unknown")
    exit()

# Negative humidity is physically impossible
assert humidity >= 0, "Telemetry Error: Negative Humidity"

# -----
# Wind Speed Input
# -----
wind_str = input("Wind Speed (km/h) : ").strip()

if wind_str:
    try:
        wind_speed = float(wind_str)
    except ValueError:
        print("\nSafety Level : Unknown")
        exit()
else:
    wind_speed = 0.0

# -----
# Heat Stress Index
# -----
hsi = temperature + (0.5 * humidity)

```

```

# Ternary expression required by assignment
risk_label = "Safe" if hsi < 30 else "Unsafe"

# -----
# Decision Logic
# -----

# 1. Freeze Alert
if temperature <= 0:
    safety_level = "FREEZE ALERT"

# 2. Critical
elif hsi > 45 or (temperature > 38 and humidity > 70):
    safety_level = "CRITICAL"

# 3. Cautionary
elif 30 <= hsi <= 45 and wind_speed < 5:
    safety_level = "CAUTIONARY"

# -----
# Bonus: Battery-Level Check
# -----
if not (battery_str := input("Battery Level (%) : ").strip()):
    print("Safety Level : Unknown")
    exit()

try:
    battery = float(battery_str)
except ValueError:
    print("Safety Level : Unknown")
    exit()

if battery < 20:
    safety_level = "CRITICAL"

elif battery > 80:
    safety_level = "OPERATIONAL"

```

```

# 4. Operational
else:
    safety_level = "OPERATIONAL"

# -----
# Final Output
# -----
print("\n===== RESULT =====")
print("Name :", user_name)
print("Temperature :", temperature, "C")
print("Humidity :", humidity, "%")
print("Wind Speed :", wind_speed, "km/h")
print("HSI :", round(hsi, 2))
print("Risk Label :", risk_label)
print("Safety Level :", safety_level)
print("=====")

```

Execution Trace

The following execution traces demonstrate the required logic paths, including an empty input case, a Critical heatwave case, and a Cautionary case adjusted using the Battery-Level bonus.

1. Empty Input

```

===== Climatic Risk Intelligence Module =====

Name : Punit Kumar
Temperature (C) :
Safety Level : Unknown

```

This demonstrates the walrus-operator empty-input validation.

2. Critical Heatwave

```

===== Climatic Risk Intelligence Module =====

Name : Punit Kumar

```

```

Temperature (C) : 32
Humidity (%) : 80
Wind Speed (km/h) : 10

===== RESULT =====
Name : Punit Kumar
Temperature : 32.0 C
Humidity : 80.0 %
Wind Speed : 10.0 km/h
HSI : 72.0
Risk Label : Unsafe
Safety Level : CRITICAL
=====

```

For this case:

$$HSI = 32 + (0.5 \times 80) = 72$$

Since the HSI is greater than 45, the environment is classified as **CRITICAL**.

3. Cautionary with Battery Adjustment

```

===== Climatic Risk Intelligence Module =====

Name : Punit Kumar
Temperature (C) : 25
Humidity (%) : 20
Wind Speed (km/h) : 3
Battery Level (%) : 10

===== RESULT =====
Name : Punit Kumar
Temperature : 25.0 C
Humidity : 20.0 %
Wind Speed : 3.0 km/h
HSI : 35.0
Risk Label : Unsafe
Safety Level : CRITICAL
=====

```

For this case:

$$HSI = 25 + (0.5 \times 20) = 35$$

The initial classification is **CAUTIONARY** because $30 \leq HSI \leq 45$ and wind speed is below 5 km/h.

The battery level is 10%, which is below 20%. Therefore, the bonus logic elevates the final safety level to **CRITICAL**.

Worked Examples Verification

The following values correspond to the four worked examples specified in the practical.

Temperature	Humidity	Wind	HSI	Expected Tier
-5	50	–	20.0	FREEZE ALERT
32	80	–	72.0	CRITICAL
25	20	3	35.0	CAUTIONARY
20	10	10	25.0	OPERATIONAL

Conclusion

The Climatic Risk Intelligence Module demonstrates a multi-signal decision system using temperature, humidity, and wind speed. It calculates the Heat Stress Index and classifies environmental conditions using the required ordered decision structure. The program also demonstrates the walrus operator, truthiness, short-circuit Boolean logic, ternary expressions, exception handling, and assertions. The Nested Intelligence Suite further extends the system by using battery level to modify a Cautionary classification. The implementation therefore demonstrates the main computational logic and programming techniques required by the practical.